

Discrete Structures

Assignment - 2

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Q1a) R is reflexive because relation on $\{a, b, c, d\}$ is reflexive if and only if $(a, a), (b, b), (c, c), (d, d) \in R$. All four are present in R so it's reflexive

b) R is NOT symmetric

Reasoning: Symmetric requires that if $(x, y) \in R$ then $(y, x) \in R$.

But in our relation:

$$\begin{array}{l|l} (a, c) \in R \text{ but } (c, a) \notin R & (a, d) \in R \text{ but } (d, a) \notin R \\ (b, a) \in R \text{ but } (a, b) \notin R & \end{array}$$

c) R is NOT antisymmetric

Reasoning: - Asymmetric requires that if $(x, y) \in R$ and $(y, x) \in R$ then $x = y$.

But in our relation this fails:

$$(b, c) \in R \text{ and } (c, b) \in R \text{ while } b \neq c$$

$$(b, d) \in R \text{ and } (d, b) \in R \text{ while } b \neq d$$

This violates antisymmetry so R is not ^{ant}symmetric

d) R is NOT transitive

Reasoning: - Transitivity requires that whenever $(x, y) \in R$ and $(y, z) \in R$ then $(x, z) \in R$

But in our relation this fails:

$$(c, b) \in R \text{ and } (b, d) \in R \text{ but } (c, d) \notin R$$

$$(c, b) \in R \text{ and } (b, a) \in R \text{ but } (c, a) \notin R$$

$$(d, b) \in R \text{ and } (b, a) \in R \text{ but } (d, a) \notin R$$

$$(d, b) \in R \text{ and } (b, c) \in R \text{ but } (d, c) \notin R$$

$$(a, d) \in R \text{ and } (d, b) \in R \text{ but } (a, b) \notin R$$

$$(a, c) \in R \text{ and } (c, b) \in R \text{ but } (a, b) \notin R$$

Q1e) R is NOT irreflexive

Reasoning: Irreflexive requires that no $(x,x) \in R$ but in our relation $(a,a), (b,b), (c,c)$ and $(d,d) \in R$. Therefore it doesn't meet irreflexive requirement.

Q1f) R is NOT asymmetric

Reasoning: Asymmetric means if $(x,y) \in R$ then $(y,x) \notin R$ (and asymmetry implies irreflexive + antisymmetry). As per previous part R is not irreflexive and not antisymmetric so it is not asymmetric too.

Q2a) $a=b$ means $(a,b) \in R \iff a=b$

Ordered pair: $\{(0,0), (1,1), (2,2), (3,3)\}$

b) $a+b=4$ means $(a,b) \in R \iff a+b=4$

sol: $b=4-a$ where $b \in B$

so Ordered Pair: $\{(1,3), (2,2), (3,1), (4,0)\}$

c) $(a,b) \in R \iff a > b$

so $a \in A, b \in B$ and $a > b$

Ordered Pair: $\{(1,0), (2,0), (2,1), (3,0), (3,1), (3,2), (4,0), (4,1), (4,2), (4,3)\}$

d) $(a,b) \in R \iff a \mid b$ (a divides b)

Ordered Pair: $\{(1,1), (1,2), (1,3), (2,0), (2,2), (3,0), (3,3), (4,0), (4,4), (1,0)\}$

e) $\text{GCD}(a,b) = 1$

for $a=0$; only $\text{GCD}(0,1)=1$ gives $(0,1)$

for $a=1$; $\text{gcd}(1,b)=1$ for every $b \in B$

for $a=2$; relatively prime to 1 and 3 only = $(2,1), (2,3)$

for $a=3$; relatively prime to 1 and 2 only $\Rightarrow (3,1), (3,2)$

for $a=4$; relatively prime to 1 and 3 only $\Rightarrow (4,1), (4,3)$

Ordered Pairs: $\{(0,1), (1,0), (1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (4,1), (4,3)\}$

Q2f) $\text{LCM}(a,b) = 2$

$\text{lcm}(1,2) = 2$

$\text{lcm}(2,1) = 2$

$\text{lcm}(2,2) = 2$

no other $a \in A, b \in B$ gives $\text{lcm} = 2$

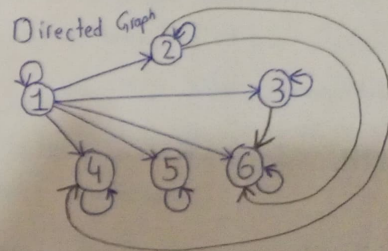
Ordered Pair = $\{(1,2), (2,1), (2,2)\}$

Q23) $R = \{(a,b) \mid a \text{ divides } b\}$ set = $\{1, 2, 3, 4, 5, 6\}$

Order Pair: $\{(1,1), (1,2), (1,3), (1,4), (1,5), (1,6), (2,2), (2,4), (2,6), (3,3), (3,6), (4,4), (5,5), (6,6)\}$

Matrix Form

a \ b	1	2	3	4	5	6
1	1	1	1	1	1	1
2	0	1	0	1	0	1
3	0	0	1	0	0	1
4	0	0	0	1	0	0
5	0	0	0	0	1	0
6	0	0	0	0	0	1



Testing if it's partial Order Relation or Equivalence Relation

• Reflexive: It is reflexive

Reasoning: for every $x \in \{1, 2, 3, 4, 5, 6\}$, $(x,x) \in R$ because x (every integer divide) itself. Hence as $(1,1), (2,2), (3,3), (4,4), (5,5), (6,6)$ exists $\therefore$ it's reflexive

• Transitive: It is transitive

Reasoning: Whenever $(x,y) \in R$ and $(y,z) \in R$ then $(x,z) \in R$

This criteria holds for this relation as each (x, z) occurs for every (x, y) and (y, z)

- Symmetric: It is NOT symmetric

Reasoning: As symmetric requires that if $(x, y) \in R$ then $(y, x) \in R$. But this doesn't happen in our case

$$\begin{array}{l|l} (1, 2) \in R \text{ but } (2, 1) \notin R & (1, 5) \in R \text{ but } (5, 1) \notin R \\ (1, 3) \in R \text{ but } (3, 1) \notin R & (1, 6) \in R \text{ but } (6, 1) \notin R \\ (1, 4) \in R \text{ but } (4, 1) \notin R & (2, 4) \in R \text{ but } (4, 2) \notin R \end{array}$$

NOTE: As symmetric test failed it means it is NOT an Equivalence Relation.

- Antisymmetric: It is Antisymmetric

Reasoning: If $(a, b) \in R$ and $(b, a) \in R$ then $a|b$ and $b|a$. For +ve integers that implies $a=b$. Concretely there is no pair of distinct $a \neq b$ for which both (a, b) and (b, a) are in R . So antisymmetry holds.

CONCLUSION:- R is a partial order relation and NOT equivalence relation.

Q4a) $\{(2, 2), (2, 3), (2, 4), (3, 2), (3, 3), (3, 4)\} = R_A$

- Reflexive: Not Reflexive

Reasoning: Not all $(x, x) \in R_A$. $(1, 1) \notin R$, $(4, 4) \notin R$ so it fails

- Symmetric: Not Symmetric

Reasoning: not all $(x, y) \in R_A$ then $(y, x) \in R_A$ condition met see $(2, 4) \in R_A$ but $(4, 2) \notin R_A$. Hence it fails

- Antisymmetric: Not Antisymmetric

Reasoning: $(2, 3) \in R_A$ and $(3, 2) \in R_A$ while $2 \neq 3$ so it violates antisymmetric criteria.

- Transitive: Yes it is Transitive

Reasoning: $(2, 3)$ & $(3, 2)$ imply $(2, 2) \in R_A$ true
 $(2, 3)$ & $(3, 4)$ imply $(2, 4) \in R_A$ true
 $(3, 2)$ & $(2, 4)$ imply $(3, 4) \in R_A$ true

All required (x, z) comes from existing (x, y) and (y, z) so it holds.

Q4b) $R_B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4)\}$

- Reflexive: Yes it is

Reasoning: All $(1, 1), (2, 2), (3, 3), (4, 4) \in R_B$

- Symmetric: Yes it is

Reasoning: Every non loop pair has its reverse: $(1, 2) \in R_B$ and $(2, 1) \in R_B$. Which fulfills criteria of $(x, y) \in R$ and $(y, x) \in R$

- Antisymmetric: Not Antisymmetric

Reasoning: It fails because $(1, 2) \in R_B$ and $(2, 1) \in R_B$ while $1 \neq 2$

- Transitive: Yes it is

Reasoning: $(1, 2)$ & $(2, 1)$ imply $(1, 1) \in R_B$ True
 $(1, 2)$ & $(2, 2)$ imply $(1, 2) \in R_B$ true
 $(2, 1)$ & $(1, 2)$ imply $(2, 2) \in R_B$ true

All required (x, z) comes from existing (x, y) and (y, z) and it itself exists so transitive holds.

Q4c) $\{(2, 4), (4, 2)\} = R_C$

- Reflexive:- No it's not

Reasoning:- Not all $(x, x) \in R_C$ such as $(1, 1), (2, 2), (3, 3), (4, 4) \notin R_C$. So it fails to hold.

• Symmetric: Yes it is

Reasoning: $(x,y) \in R$ and $(y,x) \in R$ condition proven as $(2,4)$ and $(4,2)$ are reverse of each other so symmetry holds.

• Anti Symmetric :- Not Antisymmetric

Reasoning: Since $(2,4) \in R$ and $(4,2) \in R$ while $2 \neq 4$. It violates antisymmetric condition. Hence it fails to hold

• Transitive: Not Transitive

Reasoning: As $(x,y) \in R$ and $(y,z) \in R$ then $(x,z) \in R$ condition fails here as $(2,4) \in R$ and $(4,2) \in R$ but $(2,2) \notin R$

Q4d) $R_d = \{(1,2), (2,3), (3,4)\}$

• Reflexive: No it's not

Reasoning: Not all $(x,x) \in R_d$ such as $(1,1), (2,2), (3,3), (4,4) \notin R_d$ so it fails

• Symmetric: No it's not

Reasoning: The condition of $(x,y) \in R$ and $(y,x) \in R$ is not met as $(1,2) \in R_d$ but $(2,1) \notin R_d$ | $(3,4) \in R_d$ but $(4,3) \notin R_d$
 $(2,3) \in R_d$ but $(3,2) \notin R_d$

• Antisymmetric :- Yes it is

Reasoning: No pair of (a,b) with corresponding (b,a) exist while $a \neq b$. so it's vacuously true

• Transitive: Not Transitive

Reasoning: $(1,2)$ and $(2,3) \in R_d$ but $(1,3) \notin R_d$ so it fails. Also $(2,3)$ and $(3,4) \in R_d$ but $(2,4) \notin R_d$.

Q4e) $R_e = \{(1,1), (2,2), (3,3), (4,4)\}$

• Reflexive: Yes it is

Reasoning: All loops are present. All $(x,x) \in R_e$ holds

• Symmetric :- Yes it is

Reasoning: for every $(x,y) \in R_e$ there exist $(y,x) \in R_e$. so symmetry holds

• Anti Symmetric: Yes it is

Reasoning: There are no distinct $x \neq y$ with both (x,y) and (y,x) belonging in R_e . so antisymmetry holds.

• Transitive: Yes it is

Reasoning: (x,x) composed with (x,x) gives (x,x) which is present

Q4f) $R_f = \{(1,3), (1,4), (2,3), (2,4), (3,1), (3,4)\}$

• Reflexive :- No it's not

Reasoning: No loops. $(1,1), (2,2), (3,3), (4,4) \notin R_f$ so it fails

• Symmetric: No it's not.

* Reason: Not every pair reverse exist in relation such as

$(1,4) \in R_f$ but $(4,1) \notin R_f$ | $(2,4) \in R_f$ but $(4,2) \notin R_f$
 $(2,3) \in R_f$ but $(3,2) \notin R_f$ | $(3,4) \in R_f$ but $(4,3) \notin R_f$

• Antisymmetric :- No it's not

Reason :- It's violated as $(1,3) \in R_f$ and $(3,1) \in R_f$ with $1 \neq 3$ so it doesn't hold.

• Transitive :- No it's not

Reason: $(1,3)$ and $(3,1) \in R_f$ but $(1,1) \notin R_f$
 $(2,3)$ and $(3,1) \in R_f$ but $(2,1) \notin R_f$

as we find these failure the transitive doesn't hold.

Q5a) A is taller than b

Reflexive :- NO (as no one is taller than themselves, so $(a,a) \notin R$).

Irreflexive :- YES (it means $(a,a) \notin R$ for every a , which is true as no one is taller than themselves).

Symmetric :- NO (if 'a' is taller than 'b', then 'b' is ~~not~~ taller than 'a' logic fails, cause if one's taller than other so how can the other be taller than first person).

Antisymmetric :- YES (vacuously) There are never distinct a, b with both (a,b) and (b,a) true, so the condition is vacuously satisfied.

Asymmetric :- YES (it's logically correct cause if 'a' is taller than 'b' then 'b' is NOT taller than 'a' so asymmetric holds).

Transitive :- YES (if 'a' is taller than 'b', and 'b' is taller than 'c' then 'a' is taller than 'c' so transitivity holds).

Q5b) A and B were born on the same day.

Reflexive :- YES (everyone was born on the same day as themselves so $(a,a) \in R$ for all a)

Irreflexive :- NO (as reflexive is already true so this is false)

Symmetric :- YES (if 'a' is born on the same day as 'b' then 'b' is born on the same day as 'a' $(a,b) \in R$ and $(b,a) \in R$ holds true).

Antisymmetric :- NO (as different people can be born on same day)

Asymmetric :- NO (as the relation is already symmetric)

Transitive :- YES (if 'a' and 'b' born on same day, and 'b' and 'c' born on same day, then 'a' and 'c' were born on same day so transitivity holds).

Q5c) A has the same first name as b

Reflexive :- YES (Everyone has the same first name as themselves)

Irreflexive :- NO (As reflexive condition holds, irreflexive becomes false)

Symmetric :- YES (if 'a' first name equal to 'b' first name then vice versa holds true).

Antisymmetric :- NO (as different people can have same first name)

Asymmetric :- NO (The relation is symmetric, so this condition becomes false).

Transitive :- ~~Yes~~ YES (if 'a' name is same as 'b' whose name is same as 'c'. Then 'a' name matches 'c')

Q5d) a and b have a common grandparent.

Reflexive :- YES

Reasoning :- a person has the same grandparents as themselves.

Irreflexive :- NO (as reflexive condition holds, so irreflexive is false)

Symmetric :- YES (if A & B have a common grandparent, then B & A have a common grandparent)

Antisymmetric :- NO (if A & B have a common grandparent, and B & A have a common grandparent, then it ain't necessary that A & B are the same person).

Asymmetric :- NO (The relation is symmetric, so this condition becomes false)

Transitive :- NO (if A & B have a common grandparent & B and C have a common grandparent, then it is not necessary that A and C have a common grandparent)

Q6a) Set of Whole numbers that is both symmetric and anti-symmetric

$$R = \{(a, b) \mid a = b\}$$

$$\text{eg: } R = \{(0, 0), (1, 1), (2, 2), (3, 3), \dots\}$$

Symmetric as if $(a, b) \in R$ then $(b, a) \in R$

anti-symmetric as if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$ [True]

b) Neither symmetric nor antisymmetric

$$R = \{(1, 2), (2, 1), (3, 4)\}$$

symmetric fails as $(3, 4) \in R$ but $(4, 3) \notin R$

antisymmetric fails as $(1, 2)$ and $(2, 1) \in R$ but $1 \neq 2$ fail

c) Both Symmetric and transitive

$$R = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 3), (4, 4), (5, 5), \dots\}$$

Symmetric: self-loop $(1, 1)$ $(2, 2)$ $(3, 3)$ $(4, 4)$ $(5, 5)$

reverse pair $(1, 2)$ and $(2, 1)$ $\therefore$ symmetry holds

transitive: $(1, 1)$ and $(1, 2) \in R$ imply $(1, 2) \in R$ True

$(1, 2)$ and $(2, 1) \in R$ imply $(1, 1) \in R$ True

$(2, 1)$ and $(1, 2) \in R$ imply $(2, 2) \in R$ True

$(2, 1)$ and $(1, 1) \in R$ imply $(2, 1) \in R$ True

d) Neither Symmetric nor transitive

$$R = \{(1, 2), (2, 3), (3, 1)\}$$

symmetric fails as $(1, 2) \in R$ but $(2, 1) \notin R$

$(2, 3) \in R$ but $(3, 2) \notin R$

$(3, 1) \in R$ but $(1, 3) \notin R$ fails

transitive fails as $(1, 2)$ and $(2, 3) \in R$ ~~do~~ imply $(1, 3) \notin R$

$(2, 3)$ and $(3, 1) \in R$ imply $(2, 1) \notin R$

Q7a) $R_2 \cup R_4$

$$\text{sol } R_2 = \{(1, 1), (2, 1), (3, 1), (3, 2), (2, 2), (3, 3)\}$$

$$R_4 = \{(1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$$

$$R_2 \cup R_4 = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3), (3, 1), (3, 2), (3, 3)\}$$

b) $R_3 \cup R_6$

$$\text{sol } R_3 = \{(1, 2), (1, 3), (2, 3)\}$$

$$R_6 = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

$$R_3 \cup R_6 = \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

c) $R_3 \cap R_6$

$$\text{sol } R_3 \cap R_6 = \{(1, 2), (1, 3), (2, 3)\}$$

d) $R_4 \cap R_6$

$$\text{sol } R_4 \cap R_6 = \{(1, 2), (1, 3), (2, 3)\}$$

e) $R_3 - R_6$

$$\text{sol } R_3 - R_6 = \{(1, 2), (1, 3), (2, 3)\} - \{(1, 2), (1, 3), (2, 1), (2, 3), (3, 1), (3, 2)\}$$

$$R_3 - R_6 = \emptyset$$

f) $R_6 - R_3$

$$\text{sol } R_6 - R_3 = \{(2, 1), (3, 1), (3, 2)\}$$

Q7g) $R_2 \oplus R_6$

sol

$$R_2 \oplus R_6 = (R_2 \cup R_6) - (R_2 \cap R_6)$$

$$R_2 \oplus R_6 = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\} \\ - \{(2,1), (3,1), (3,2)\}$$

$$R_2 \oplus R_6 = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$$

Q7h) $R_3 \oplus R_5$

sol

$$R_3 \oplus R_5 = (R_3 \cup R_5) - (R_3 \cap R_5)$$

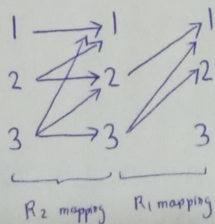
$$R_3 \oplus R_5 = \{(1,2), (1,3), (2,3), (1,1), (2,2), (3,3)\} - \{\emptyset\}$$

$$R_3 \oplus R_5 = \{(1,2), (1,3), (1,1), (2,2), (2,3), (3,3)\}$$

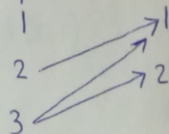
Q7i) $R_2 \circ R_1$

$$R_2 = \{(1,1), (2,1), (2,2), (3,1), (3,2), (3,3)\}$$

$$R_1 = \{(2,1), (3,1), (3,2)\}$$



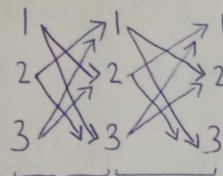
composition



$$\therefore R_2 \circ R_1 = \{(2,1), (3,1), (3,2)\}$$

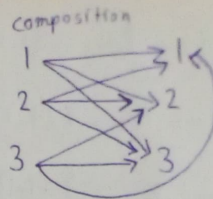
Q7j) $R_6 \circ R_6$

$$R_6 = \{(1,2), (1,3), (2,1), (2,3), (3,1), (3,2)\}$$



first R_6 map second R_6 map

$$\therefore R_6 \circ R_6 = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$$



Q8ai) $R = \{(1,1), (1,2), (1,3)\}$

Matrix =

	1	2	3
1	1	1	1
2	0	0	0
3	0	0	0

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

ii) $R = \{(1,2), (2,1), (2,2), (3,3)\}$

Matrix =

	1	2	3
1	0	1	0
2	1	1	0
3	0	0	1

$$M = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

iii) $R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$

Matrix =

	1	2	3
1	1	1	1
2	0	1	1
3	0	0	1

$$M = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

iv) $R = \{(1,3), (3,1)\}$

Matrix =

	1	2	3
1	0	0	1
2	0	0	0
3	1	0	0

$$M = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$

Q8bi) Ordered Pair = $\{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

ii) Ordered Pair = $\{(1,2), (2,2), (3,2)\}$

iii) Ordered Pair = $\{(1,1), (1,2), (1,3), (2,1), (2,3), (3,1), (3,2), (3,3)\}$

Q9a) Reflexive : Yes

Reasoning :- for any string a , $L(a) = L(a)$ thus $(a,a) \in R$ for every a . So R is reflexive

Symmetric : Yes

Reasoning : If $(a,b) \in R$ then $L(a) = L(b)$. ^{equality} which is symmetric so $L(b) = L(a)$ hence $(b,a) \in R$. Therefore symmetric holds. $(aRb = bRa)$

Transitive : Yes

Reasoning : If $(a,b) \in R$ and $(b,c) \in R$ then $L(a) = L(b)$ and $L(b) = L(c)$. $\therefore$ By transitivity of equality $L(a) = L(c)$ Therefore $(a,c) \in R$

Now as R is reflexive, symmetric, and transitive, R is an equivalence relation.

Q9b) Reflexivity : Yes

Reasoning :- for any integer a , we have $a \equiv a \pmod{m}$ because $a-a=0$ is divisible by m . Therefore $(a,a) \in R$ for all a .

Symmetry : Yes

Reasoning :- if $(a,b) \in R$, then $a \equiv b \pmod{m}$. This means that m divides $a-b$. Since m also divides $(b-a)$ we have $b \equiv a \pmod{m}$. Therefore $(b,a) \in R$.

Transitive : Yes

Reasoning : If $a \equiv b \pmod{m}$ and $b \equiv c \pmod{m}$, then $m \mid (a-b)$ and $m \mid (b-c)$. Then $m \mid [(a-b) + (b-c)] = a-c$. Therefore $a \equiv c \pmod{m}$ and $(a,c) \in R$ so transitive.

Now as R is reflexive, symmetric, and transitive, R is an equivalence relation.

Q10ai) $2^n - 1$ where $n > 0$

first 5 terms : $2^1 - 1 = 1$

$$2^2 - 1 = 3$$

$$2^3 - 1 = 7$$

$$2^4 - 1 = 15$$

$$2^5 - 1 = 31$$

$$\text{Ans} = 1, 3, 7, 15, 31$$

aii) $10 - \frac{3}{2}n$ where $n > 0$

first 5 terms : $10 - \frac{3}{2}(1) = \frac{17}{2}$

$$10 - \frac{3}{2}(2) = 7$$

$$10 - \frac{3}{2}(3) = \frac{11}{2}$$

$$10 - \frac{3}{2}(4) = 4$$

$$10 - \frac{3}{2}(5) = \frac{5}{2}$$

$$\text{Ans} = \frac{17}{2}, 7, \frac{11}{2}, 4, \frac{5}{2}$$

aiii) $\frac{(-1)^n}{n^2}$ where $n > 0$

first 5 terms : $\frac{(-1)^1}{1^2} = -1$

$$\frac{(-1)^2}{2^2} = \frac{1}{4}$$

$$\frac{(-1)^3}{3^2} = -\frac{1}{9}$$

$$\frac{(-1)^4}{4^2} = \frac{1}{16}$$

$$\frac{(-1)^5}{5^2} = -\frac{1}{25}$$

$$\text{Ans} = -1, \frac{1}{4}, -\frac{1}{9}, \frac{1}{16}, -\frac{1}{25}$$

iv) $\frac{3n+4}{2n-1}$ where $n > 0$

first 5 terms: $\frac{3(1)+4}{2(1)-1} = \frac{7}{1}$
 $\frac{3(2)+4}{2(2)-1} = \frac{10}{3}$
 $\frac{3(3)+4}{2(3)-1} = \frac{13}{5}$
 $\frac{3(4)+4}{2(4)-1} = \frac{16}{7}$
 $\frac{3(5)+4}{2(5)-1} = \frac{19}{9}$
Ans = $7, \frac{10}{3}, \frac{13}{5}, \frac{16}{7}, \frac{19}{9}$

Q10bi) Sequence = -15, -22, -29, -36, ...; 11th term

step 1: Check difference ($a_n - a_1$)

$$-22 - (-15) = -7$$

$$-29 - (-22) = -7$$

$$-36 - (-29) = -7$$

Common difference (d) = -7

∴ It's Arithmetic Sequence

step 2: Finding 11th Term

$$a_n = a_1 + (n-1)d$$

$$a_{11} = a_1 + (n-1)d$$

$$a_{11} = -15 + (11-1)(-7)$$

$$a_{11} = -85 \text{ is the 11th term of the sequence} \text{ Ans}$$

Data
 $a_1 = -15$
 $d = -7$
 $n = 11$

bii) Sequence = $a-42b, a-39b, a-36b, a-33b, \dots$; 15th term

Step 1: Check Difference ($T_n - T_n$)

$$(a-39b) - (a-42b) = 3b$$

$$(a-36b) - (a-39b) = 3b$$

$$(a-33b) - (a-36b) = 3b$$

Common Difference (d) = 3b

∴ It's Arithmetic Sequence

Step 2: Finding 15th Term

$$a_n = a_1 + (n-1)d$$

$$a_{15} = (a-42b) + (15-1)(3b)$$

$$a_{15} = a - 42b + 42b$$

$$a_{15} = a \text{ is the 15th term of the sequence} \text{ Ans}$$

Data
 $a_1 = a-42b$
 $n = 15$
 $d = 3b$

Q10iii) Sequence = $4, 3, \frac{9}{4}, \dots$; 17th term

step 1: Check ratio ($r = \frac{a_n}{a_{n-1}}$)

$$\frac{a_2}{a_{2-1}} = \frac{3}{4}$$

so common ratio (r) = $\frac{3}{4}$

$$\frac{a_3}{a_{3-1}} = \frac{(9/4)}{3} = \frac{3}{4}$$

∴ It's Geometric Sequence

step 2: Finding 17th Term

$$a_n = a_1 r^{n-1}$$

$$a_{17} = 4 \left[\frac{3}{4} \right]^{17-1}$$

Data
 $a_1 = 4$
 $r = \frac{3}{4}$
 $n = 17$

$$a_{17} = 4 \left[\frac{3}{4} \right]^{16} \text{ or } 0.04009 \text{ is the 17th term of the sequence} \text{ Ans}$$

biv) Sequence = $32, 16, 8, \dots$; 9th term

step 1: Check ratio ($r = \frac{a_n}{a_{n-1}}$)

$$\frac{a_2}{a_{2-1}} = \frac{16}{32} = \frac{1}{2}$$

so common ratio (r) = $\frac{1}{2}$

$$\frac{a_3}{a_{3-1}} = \frac{8}{16} = \frac{1}{2}$$

∴ It's Geometric Sequence

step 2: Finding 9th term

$$a_n = a_1 r^{n-1}$$

$$a_9 = 32 \left(\frac{1}{2} \right)^{9-1}$$

$$a_9 = \frac{32}{2^8}$$

$$a_9 = \frac{1}{8} \text{ Ans as it's the 9th term of the sequence}$$

Data
 $a_1 = 32$
 $n = 9$
 $r = \frac{1}{2}$

Q11a i) $T_3 = 10$ and $T_5 = 2\frac{1}{2}$

$$\begin{array}{l|l} T_3 = ar^2 & T_5 = ar^4 \\ 10 = ar^2 & \frac{5}{2} = ar^4 \end{array}$$

Divide both eqs

$$\frac{T_5}{T_3} \Rightarrow \frac{ar^4}{ar^2} = \frac{(5/2)}{10}$$

$$\sqrt{r^2} = \sqrt{\frac{1}{4}}$$

$$r = \pm \frac{1}{2}$$

~~Divide both eqs~~

$$\frac{T_5}{T_3} = \frac{10}{(5/2)}$$

$$\frac{ar^4}{ar^2} = 4$$

$$\sqrt{r^2} = \sqrt{4}$$

$$r = \pm 2$$

when $r = +\frac{1}{2}$

$$\begin{array}{l} T_3 = ar^2 \\ 10 = a \left(\frac{1}{2}\right)^2 \\ 40 = a \end{array}$$

so GP = 40, 20, 10, 5, $\frac{5}{2}, \dots$

$$\therefore GP = 40 \left(\frac{1}{2}\right)^{n-1}$$

aii) $T_5 = 8$ and $T_8 = -\frac{64}{27}$

$$\begin{array}{l|l} T_5 = ar^4 & T_8 = ar^7 \\ 8 = ar^4 & -\frac{64}{27} = ar^7 \end{array}$$

when $r = -\frac{2}{3}$

$$\begin{array}{l} T_5 = ar^4 \\ 8 = a \left(-\frac{2}{3}\right)^4 \\ \frac{81}{2} = a \end{array}$$

so GP $\Rightarrow \frac{81}{2} \left(-\frac{2}{3}\right)^n = T_n$

Divide both eqs

$$\frac{T_8}{T_5} \Rightarrow \frac{ar^7}{ar^4} = \frac{(-64/27)}{8}$$

$$r^3 = -\frac{8}{27}$$

$$r = -\frac{2}{3}$$

Sequence = $\frac{81}{2}, -27, 18, -12, \dots$

$T_5 = \frac{81}{2}, -\frac{16}{3}, \frac{32}{9}, -\frac{64}{27}, \dots$

Q11bi) $T_4 = 7$ and $T_{16} = 31$

$$\begin{array}{l|l} T_4 = a + (n-1)d & T_{16} = a + (n-1)d \\ 7 = a + 3d & 31 = a + 15d \\ \text{eq 1} & \text{eq 2} \end{array}$$

AP $\Rightarrow T_n = a + (n-1)d$

$$T_n = 1 + 2(n-1)$$

Sequence = 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, $\dots$

T_4 T_{16}

subtract eq 1 from eq 2

$$(a+3d) - (a+15d) = 7-31$$

$$-12d = -24$$

$$d = 2$$

for a

$$\begin{array}{l} 7 = a + 3d \\ 7 = a + 6 \\ a = 1 \end{array}$$

Q11bii) $T_5 = 86$ and $T_{10} = 146$

$$\begin{array}{l|l} T_5 = a + (n-1)d & T_{10} = a + (n-1)d \\ 86 = a + 4d & 146 = a + 9d \\ \text{eq 1} & \text{eq 2} \end{array}$$

AP $\Rightarrow T_n = a + (n-1)d$

$$T_n = 38 + 12(n-1)$$

Sequence = 38, 50, 62, 74, 86, 98, 110, 122, 134, 146, $\dots$

T_5 T_{10}

subtract eq 1 from eq 2

$$(a+4d) - (a+9d) = 86-146$$

$$-5d = -60$$

$$d = 12$$

for a

$$\begin{array}{l} 86 = a + 4d \\ 86 - 4(12) = a \\ 38 = a \end{array}$$

Q12a) Qty of numbers between 256 and 784 that are divisible by 7.

common difference (d) = 7

starting term (a) = 259 ; as $259/7 = 37$

Last term (T_n) = 784 ; as $784/7 = 112$

so $T_n = a + (n-1)d$

$$784 = 259 + 7(n-1)$$

$$75 + 1 = n$$

$$76 = n$$

$\therefore$ There are 76 numbers between 256 and 784 that's divisible by 7.

Sum of those 76 numbers = ?

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

$$S_{76} = \frac{76}{2} [2(259) + (76-1)(7)]$$

$$S_{76} = 39634 \text{ Ans}$$

Q12b) $T_1 = \frac{1}{n}$ $T_n = \frac{n^2-n+1}{n}$

Since: $S_n = \frac{n}{2} [a + l]$
 $\downarrow$ first term $\quad \downarrow$ last term

$$S_n = \frac{n}{2} \left[\frac{1}{n} + \frac{n^2-n+1}{n} \right]$$

$$S_n = \frac{n}{2} \left[\frac{n^2-n+2}{n} \right]$$

$$S_n = \frac{n^2-n+2}{2} \text{ Ans}$$

OR use first formula

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

for d

$$T_n = a + (n-1)d$$

$$\frac{n^2-n+1}{n} = \frac{1}{n} + (n-1)d$$

$$d = 1$$

$$\text{so } S_n = \frac{n}{2} \left[\frac{2}{n} + n-1 \right]$$

$$S_n = \frac{n^2-n+2}{2}$$

same result.

Q13a) Summation Notation: $\sum_{j=1}^{100} (1/j)$

Q13bi) $\sum_{k=4}^8 (-1)^k = (-1)^4 + (-1)^5 + (-1)^6 + (-1)^7 + (-1)^8$

$$= 1 - 1 + 1 - 1 + 1$$

$$= 1 \text{ Ans}$$

13bii) $\sum_{j=1}^5 (j)^2 = (1)^2 + (2)^2 + (3)^2 + (4)^2 + (5)^2$

$$= 1 + 4 + 9 + 16 + 25$$

$$= 55 \text{ Ans}$$

Q14a) Sequence = 40, 33, 26, ...

step 1: check if it's AP

find common difference ($d = T_{n+1} - T_n$)

$$d = T_2 - T_1 = 33 - 40 = -7$$

$$d = T_3 - T_2 = 26 - 33 = -7$$

so common difference (d) = -7

$$\therefore \text{AP} \Rightarrow T_n = a + (n-1)d$$

$$T_n = 40 + (n-1)(-7)$$

for distance dropped in 5th second

$$T_n = 40 + (n-1)d$$

$$T_5 = 40 + (5-1)(-7)$$

$$T_5 = 40 + (4)(-7)$$

$$T_5 = 12 \text{ ft dropped in 5th second Ans}$$

Total Distance covered by the ball after 5th second = ?

$$S_5 = \frac{n}{2} [2a + (n-1)d]$$

$$S_5 = \frac{5}{2} [2(40) + (4)(-7)]$$

$$S_5 = 130 \text{ ft}$$

$$\therefore \text{Total Drop in 5 seconds is 130 ft.}$$

Q14b) Data

Annual Growth rate = 7% (means multiplying by 1.07 every year)

Present Population (P_0) = 70000

after 6 year = ?

$$P_6 = P_0 \times r^6$$

$$P_6 = (70000)(1.07)^6$$

$$P_6 = 105051.1246$$

$$P_6 \approx 105051 \text{ people population after 6yrs}$$

Q15ai) $a_n = -2a_{n-1}$, $a_0 = -1$

Sol

$$a_0 = -1$$

$$a_1 = -2a_0 = -2(-1) = 2$$

$$a_2 = -2a_1 = -2(2) = -4$$

$$a_3 = -2a_2 = -2(-4) = 8$$

$$a_4 = -2a_3 = -2(8) = -16$$

$$a_5 = -2a_4 = -2(-16) = 32$$

∴ First six Terms are
-1, 2, -4, 8, -16, 32

aii) $a_n = a_{n-1} - a_{n-2}$, $a_0 = 2$ $a_1 = -1$

Sol

$$a_0 = 2$$

$$a_1 = -1$$

$$a_2 = a_1 - a_0 = -1 - 2 = -3$$

$$a_3 = a_2 - a_1 = -3 - (-1) = -2$$

$$a_4 = a_3 - a_2 = -2 - (-3) = 1$$

$$a_5 = a_4 - a_3 = 1 - (-2) = 3$$

∴ First six terms are
2, -1, -3, -2, 1, 3.

aiii) $a_n = 3a_{n-1}^2$, $a_0 = 1$

Sol

$$a_0 = 1$$

$$a_1 = 3a_0^2 = 3(1)^2 = 3$$

$$a_2 = 3a_1^2 = 3(3)^2 = 27$$

$$a_3 = 3a_2^2 = 3(27)^2 = 2187$$

$$a_4 = 3a_3^2 = 3(2187)^2 = 14348907$$

$$a_5 = 3a_4^2 = 3(14348907)^2 = 6.176733963 \times 10^{14}$$

∴ First six terms are :- 1, 3, 27, 2187, 14348907, 6.18×10^{14}

Q15aiv) $a_n = na_{n-1} + a_{n-2}^2$, $a_0 = -1$, $a_1 = 0$

Sol

$$a_0 = -1$$

$$a_1 = 0$$

$$a_2 = 2a_1 + a_0^2 = 2(0) + (-1)^2 = 1$$

$$a_3 = 3a_2 + a_1^2 = 3(1) + (0)^2 = 3$$

$$a_4 = 4a_3 + a_2^2 = 4(3) + (1)^2 = 13$$

$$a_5 = 5a_4 + a_3^2 = 5(13) + (3)^2 = 74$$

∴ First six terms are : -1, 0, 1, 3, 13, 74

Q15bi) $(-2)^n = a_n$

$$a_0 = (-2)^0 = 1$$

$$a_1 = (-2)^1 = -2$$

$$a_2 = (-2)^2 = 4$$

$$a_3 = (-2)^3 = -8$$

∴ $a_0 = 1$, $a_1 = -2$
 $a_2 = 4$, $a_3 = -8$

bii) $a_n = 3$

$$a_0 = 3$$

$$a_1 = 3$$

$$a_2 = 3$$

$$a_3 = 3$$

} constant sequence

biii) $a_n = 7 + 4^n$

$$a_0 = 7 + 4^0 = 8$$

$$a_1 = 7 + 4^1 = 11$$

$$a_2 = 7 + 4^2 = 23$$

$$a_3 = 7 + 4^3 = 71$$

∴ (8, 11, 23, 71) respectively for (a_0, a_1, a_2, a_3)

biv) $a_n = 2^n + (-2)^n$

$$a_0 = 2^0 + (-2)^0 = 1 + 1 = 2$$

$$a_1 = 2^1 + (-2)^1 = 2 - 2 = 0$$

$$a_2 = 2^2 + (-2)^2 = 4 + 4 = 8$$

$$a_3 = 2^3 + (-2)^3 = 8 - 8 = 0$$

∴ 2, 0, 8, 0 Ans

Q16i) For Graph (i)

- It has undirected edges
- It has multiple edges
- It has no loops

CONCLUSION: It is an UNDIRECTED multigraph.

ii) For Graph (ii)

- It has undirected edges
- It has no multiple edges
- It has no loops

CONCLUSION: It is an UNDIRECTED simple graph.

iii) For Graph (iii)

- It has undirected edges
- It has multiple edges
- It has loops (3)

CONCLUSION: It is an UNDIRECTED pseudo graph

iv) For Graph (iv)

- It has directed edges
- It has multiple edges
- It has two loops

CONCLUSION: It is an DIRECTED multigraph.

Q17a) $A_1 = \{0, 2, 4, 6, 8\}$

$A_2 = \{0, 1, 2, 3, 4\}$

$A_3 = \{1, 3, 5, 7, 9\}$

$A_4 = \{5, 6, 7, 8, 9\}$

$A_5 = \{0, 1, 8, 9\}$

working

$A_1 \cap A_2 = \{0, 2, 4\}$ edge

$A_1 \cap A_3 = \emptyset$ no edge

$A_1 \cap A_4 = \{6, 8\}$ edge

$A_1 \cap A_5 = \{0, 8\}$ edge

$A_2 \cap A_3 = \{1, 3\}$ edge

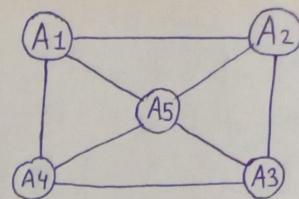
$A_2 \cap A_4 = \emptyset$ no edge

$A_2 \cap A_5 = \{0, 1\}$

$A_3 \cap A_4 = \{5, 7, 9\}$ edge

$A_3 \cap A_5 = \{1, 9\}$ edge

$A_4 \cap A_5 = \{8, 9\}$ edge



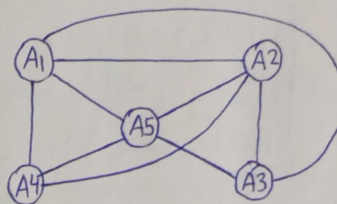
Q17b) $A_1 = \{\dots, -4, -3, -2, -1, 0\}$

$A_2 = \{\dots, -2, -1, 0, 1, 2, \dots\}$

$A_3 = \{\dots, -6, -4, -2, 0, 2, 4, 6\}$

$A_4 = \{\dots, -5, -3, -1, 1, 3, 5, \dots\}$

$A_5 = \{\dots, -6, -3, 0, 3, 6, \dots\}$



Q18ai) For Graph (i)

No. of Vertices: 5

No. of Edges: 13

$\deg(a) = 6$

$\deg(b) = 6$

$\deg(c) = 6$

$\deg(d) = 5$

$\deg(e) = 3$

Neighbour hood Vertices

$N(a) = \{a, b, e\}$

$N(b) = \{a, c, d, e\}$

$N(c) = \{b, c, d\}$

$N(d) = \{b, c, e\}$

$N(e) = \{a, b, d\}$

Q18aii For Graph (ii)

No. of Vertices :- 9

No. of Edges :- 12

$\deg(a) = 3$

$\deg(e) = 6$

$\deg(i) = 3$

$\deg(b) = 2$

$\deg(f) = 0$

$\deg(c) = 4$

$\deg(g) = 4$

$\deg(d) = 0$

$\deg(h) = 2$

Neighbourhood vertices

$N(a) = \{c, e, i\}$

$N(e) = \{a, b, c, g\}$

$N(b) = \{e, h\}$

$N(f) = \emptyset$ (no vertices)

$N(c) = \{a, e, g, i\}$

$N(g) = \{c, e\}$

$N(d) = \emptyset$ (no vertices)

$N(h) = \{b, i\}$

$N(i) = \{a, c, h\}$

Q18bi For Graph (i)

No. of vertices :- 5

No. of edges :- 13

In-degree of vertices

$d^-(a) = 6$

$d^-(c) = 2$

$d^-(b) = 1$

$d^-(d) = 4$

$d^-(e) = 0$

Out-degree vertices

$d^+(a) = 1$

$d^+(d) = 2$

$d^+(b) = 5$

$d^+(e) = 0$

$d^+(c) = 5$

18bii For Graph (ii)

No. of vertices = 4

No. of edges = 8

In-degree of vertex

$d^-(a) = 2$

$d^-(c) = 2$

$d^-(b) = 3$

$d^-(d) = 1$

Out-degree of vertex

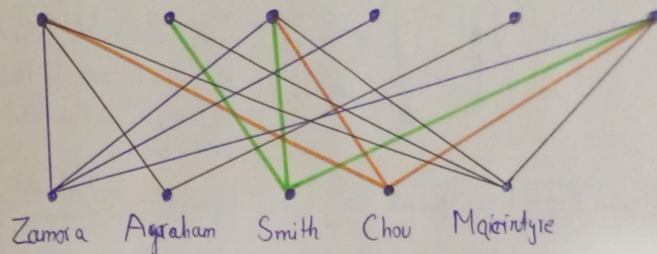
$d^+(a) = 2$

$d^+(d) = 1$

$d^+(b) = 4$

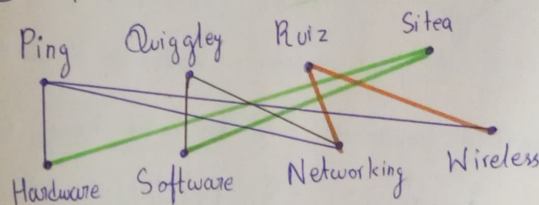
$d^+(c) = 1$

Q19a Planning Publicity Sales Marketing Development Industry Relation



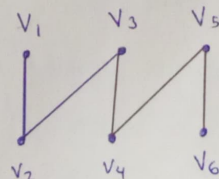
The above graph is a Bipartite Graph

Q19b Bipartite Graph

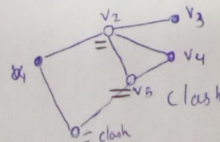


Q20a NOT a Bipartite graph (as 'c' is adjacent to 'a' and 'd')

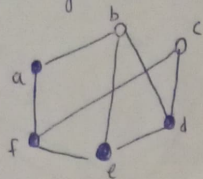
b) It is a Bipartite graph ($A = \{v_1, v_3, v_5\}$ and $B = \{v_2, v_4, v_6\}$)



c) NOT a Bipartite graph (as ' v_2 ' and ' v_4 ' have color clash of white if we paint vertex black and white (if they are adjacent to each other.) (as ' v_2 ' is adjacent to ' v_4, v_5 ')



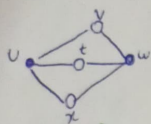
Q20d) Doing color algorithm test



color clash at f, e, and d as graph cause them to be black and white at same time

NOT a Bipartite graph (as b is adjacent to 'd' and 'e')

Q21ai) It's a Bipartite Graph as it passes color theory test



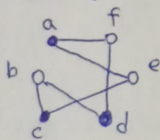
$$V_1 = \{u, w\}$$

$$V_2 = \{v, t, x\}$$



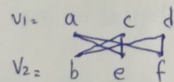
so that every edge of graph connects a vertex in V_1 and V_2

aii) It's a Bipartite graph as it passes color theory test



$$V_1 = \{a, c, d\}$$

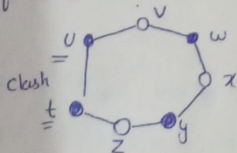
$$V_2 = \{b, e, f\}$$



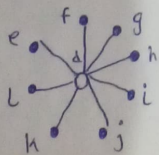
see bipartite

so we see that every edge of graph connects a vertex in V_1, V_2

aiii) Not a Bipartite graph as u and t are adjacent and of same color.



aiv) It's a Bipartite graph as it passes color theory test

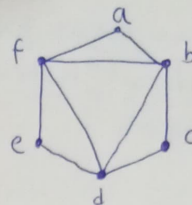


$$V_1 = \{f, g, h, i, j, k, l, e\}$$

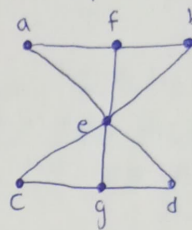
$$V_2 = \{d\}$$

we see that every edge of graph connects a vertex in V_1, V_2

Q21bi) Union of Graph



bii) Union of Graph



Q22a) Data:

Vertices = 4 assume (a, b, c, d to be vertices name)
deg(a) = 1, deg(b) = 1, deg(c) = 2, deg(d) = 3

sol:

$$\text{Total degree of graph} = 1 + 1 + 2 + 3 = 7 \text{ (odd)}$$

According to Handshaking Theorem, total degree of graph should be even.

Therefore graph is not possible for this property.

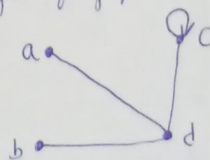
Q22bi) Data:

Vertices = 4 assume (a, b, c, d to be vertices name)
deg(a) = 1, deg(b) = 1, deg(c) = 3, deg(d) = 3

sol:

$$\text{Total degree of graph} = 1 + 1 + 3 + 3 = 8 \text{ (even)}$$

∴



Q22c) A simple graph with vertices of degrees 1, 1, 3, 3

Sol It does not exist. As we can see from previous part an 'Undirected Pseudograph' is formed due to loop on vertex c. ~~which is~~

Q23a) No this is not possible

Sol :- As per Handshaking Theorem

$$\text{Total degree} = 15 \times 3 = 45 \text{ (odd)}$$

so $45 \neq 2e$ as we get decimal

Conclusion :- Since total degree of the graph is odd, it is not possible to make the graph such that each 15 vertices have degree = 3

23b) Yes this is possible

Sol :- As per Handshaking Theorem

$$\text{Total degree} = 4 \times 3 = 12 \text{ (even)}$$

so $12 = 2e$ (possible as we get int)

Conclusion :- Since total degree of the graph is ~~odd~~ even, it is possible to make a graph such that each 4 vertices have degree 3.

23c) Number of Handshake = $C(10, 2) = \frac{10!}{2!(10-2)!} = 45$

The scenario is an example of a complete graph in graph theory specifically K_{10} , where each person is a vertex connected to every other vertex by an edge (handshake) which is unique.

23d) Being friends with exactly 2 people is possible;

$$\text{Total degree} = 5 \times 2 = 10 \text{ (even)}$$

so $10 = 2e$ (possible as we get integer)

Conclusion :- This forms a cycle graph (pentagon) where each person is friends with two neighbors

23d) Being friends with exactly 3 people is NOT possible

$$\text{Total degree} = 5 \times 3 = 15 \text{ (odd)}$$

so $15 \neq 2e$ (not possible as we get decimal)
 $7.5 = e \rightarrow$

Conclusion :- Since total degree of the graph is odd, handshaking theorem fails and we can't make a graph such that each 5 vertices have degree 3.

Q23e) Handshaking theorem

$$\text{Total degree} = 2 \text{ (edges)}$$

$$\text{Vertices} \times \text{degree} = 2 \text{ (edges)}$$

$$\text{Vertices} (4) = 2(10)$$

$$\text{Vertices} = \frac{20}{4}$$

$$\text{Vertices} = 5$$

Data
degree = 4
no. of edge = 10

Q24) For Graph (W)

a) ~~Graph (W) is connected because there is a path between any pair of its vertices~~

b)

Q24a) • Graph (W) is connected because there is a path between any pair of its vertices.

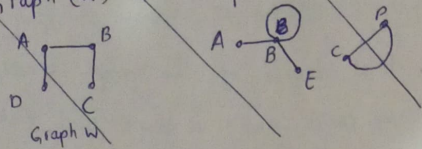
• Graph (Y) is not connected because there is no path from connected component $\{A, D, E\}$ to connected component $\{B, C\}$

• Graph (X) is connected because there is a path between any pair of its vertices.

• Graph X is not connected because there is no path between its connected component $\{A, B, E\}$ and connected component $\{C, D\}$

~~Q24a) Graph (W), Graph (V) and Graph (Z) are free of cycles/loops~~

Q24b) Graph (W) and Graph (X) are cycle free



Q24b) Graph (W) is cycle free only

Q24c) Graph (W), Graph (Y) and Graph (Z) are loop free

Q24d) Graph (W) and Graph (Y) are simple graph

Q25a) Data:

Vertices $(G) = 5$
edges = 7

Vertices $(G') = 5$
edges = 7

degree sequence $\Rightarrow$

4, 2, 3, 3, 2
 v_1, v_2, v_3, v_4, v_5

3, 4, 2, 2, 3
 w_1, w_2, w_3, w_4, w_5

As the conditions are satisfied therefore it's ~~an~~ isomorphic

$g: V(G) \rightarrow V(G') \Rightarrow$

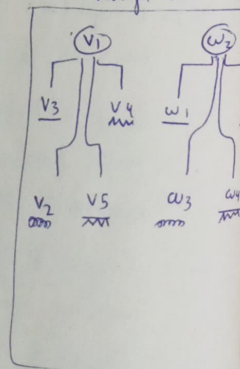
$$g(v_1) = w_2$$

$$g(v_2) = w_3$$

$$g(v_3) = w_1$$

$$g(v_4) = w_5$$

$$g(v_5) = w_4$$



Q25b) Data:

Vertices $(G) = 6$

edges $(G) = 7$

Vertices $(G') = 6$

edges $(G') = 7$

Both graphs have 6 vertices and 7 edges.

Now for degree sequence

G degree = 3, 3, 3, 2, 1, 2
 $v_1, v_2, v_3, v_4, v_5, v_6$

G' degree = 1, 3, 2, 3, 3, 2
 $u_1, u_2, u_3, u_4, u_5, u_6$

Both graphs have same amount of degree qty.

Conclusion:- They are isomorphic

$g: V(G) \rightarrow V(G') \Rightarrow$

$$g(v_1) = u_5$$

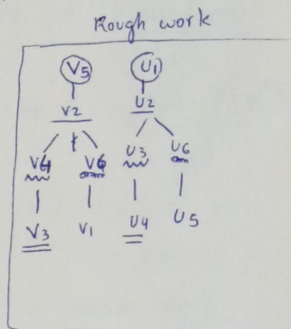
$$g(v_2) = u_2$$

$$g(v_3) = u_4$$

$$g(v_4) = u_3$$

$$g(v_5) = u_1$$

$$g(v_6) = u_6$$



Q25c) Data:

Vertices $(G) = 7$

Edges $(G) = 9$

Vertices $(G') = 7$

Edges $(G') = 9$

Degree Sequence $\Rightarrow$

2, 3, 3, 2, 2, 4, 2
 $v_1, v_2, v_3, v_4, v_5, v_6, v_7$

Degree Sequence $\Rightarrow$

4, 2, 3, 3, 2, 2, 2
 $u_1, u_2, u_3, u_4, u_5, u_6, u_7$

All conditions are satisfied

Conclusion:- They are isomorphic

$g: V(G) \rightarrow V(G') \Rightarrow$

$$g(v_1) = u_5$$

$$g(v_2) = u_4$$

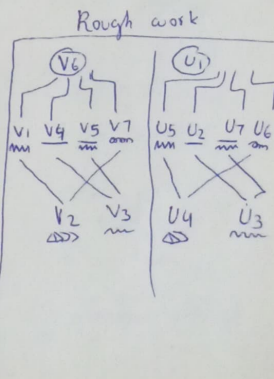
$$g(v_3) = u_3$$

$$g(v_4) = u_2$$

$$g(v_5) = u_7$$

$$g(v_6) = u_1$$

$$g(v_7) = u_6$$



As the conditions are met. It's isomorphic

see mapping

$$\begin{aligned} u_1 &= v_1 = t_1 = w_1 \\ u_2 &= v_2 = t_2 = w_2 \\ u_3 &= v_3 = t_3 = w_3 \end{aligned} \quad \left. \begin{aligned} u_4 &= v_4 = t_4 = w_4 \\ u_5 &= v_5 = t_5 = w_5 \end{aligned} \right\}$$

Q27a) Shortest path = abehlmps (16)

	N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
a		2,a	4,a	7,a	∞	∞	∞	∞
ad		4,a	4,a	∞	∞	∞	∞	∞
adb		4,a	3,b	6,c	5,d	∞	∞	∞
adbe		4,a	6,c	6,c	6,e	8,f	10,f	10,f
adbec		6,c	5,d	6,e	7,g	7,h	10,i	10,i
adbecg		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfh		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhkl		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhkli		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklir		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirm		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmp		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpq		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpqr		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpqrst		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpqrstv		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpqrstvw		6,c	6,e	7,g	7,h	10,i	10,i	10,i
adbecgfhklirmnpqrstvwxyz		6,c	6,e	7,g	7,h	10,i	10,i	10,i

	N	D(b)	D(c)	D(d)	D(e)	D(f)	D(g)	D(z)
a		4,a	3,a					
ac		4,a		6,c	9,c			
acb				6,c	9,c			
acbd					7,d	11,d		
acbde						11,d	12,e	
acbdfe							12,e	18,f
acbdfez								16,g

shortest path = acdegz (16)

Q27c)

	N	D(b)	D(c)	D(d)	D(e)	D(f)	D(z)
a		12,a	10,a			12,a	
ac		12,a		21,c	13,c	12,a	
acb				21,c	13,c	21,f	
acbf					13,c		21,f
acbfz							19,e

shortest path = acez (19)

Q28a) sol

$$ABCOA = 30 + 30 + 25 + 40 = 125$$

$$ABDCA = 30 + 35 + 25 + 50 = 140$$

$$ACBDA = 50 + 30 + 35 + 40 = 155$$

Route that will minimize the total distance traveled is A-B-C-D-A. its distance is 125

Q28b) sol

(Hamilton Circuit)

$$ABCD A = 20 + 30 + 12 + 35 = 97$$

$$ABDCA = 20 + 34 + 12 + 42 = 108$$

$$ACBDA = 42 + 30 + 34 + 35 = 141$$

$$ACDBA = 42 + 12 + 34 + 20 = 108$$

$$ADBCA = 35 + 34 + 30 + 42 = 141$$

$$ADCBA = 35 + 12 + 30 + 20 = 97$$

These are same as above 3 so we can ignore them

Route that will minimize total distance is A-B-C-D-A.

As it's 97 distance, (also ADCBA give 97)

Q28c) sol

Hamilton circuit

$$ADBCA = 20 + 25 + 35 + 15 = 95$$

$$ABCD A = 10 + 35 + 30 + 20 = 95$$

$$ACBDA = 15 + 35 + 25 + 20 = 95$$

$$ACDBA = 15 + 30 + 25 + 10 = 80 \text{ min}$$

$$ABDCA = 10 + 25 + 30 + 15 = 80 \text{ min}$$

$$ADCBA = 20 + 30 + 35 + 10 = 95$$

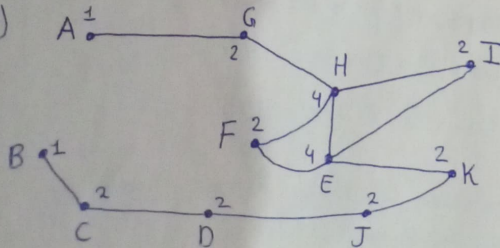
same route travel

Route that will minimize total distance is A-C-D-B-A as it's distance is 80

Q29a) It is indeed possible to move from room A and exit through room E using all other rooms door only once.

Route :- A → H → G → B → C → D → G → F → E

Q29b)



$d(A) = d(B) = 1$ which is odd

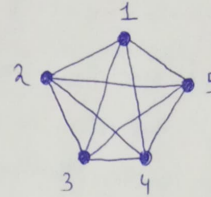
$d(C) = d(D) = d(E) = d(F) = d(G) = d(H) = d(I) = d(J) = d(K)$

these are even

Hence it's an Euler Path

Route :- A → G → H → F → E → I → H → E → K → J → D → C

Q29c)



$\deg(1) = 4$
 $\deg(2) = 4$
 $\deg(3) = 4$
 $\deg(4) = 4$
 $\deg(5) = 4$

Yes it is an Euler path as Graph is connected and each vertex has an even degree of 4, and only initial and final vertex have odd degree. (0 all vertices have odd deg)

Since the graph is connected and all vertices have even degree, an Eulerian circuit exists.

This means it is possible to arrange the domino tiles in a connected series such that the touching ends have the same number, and the series will use all tiles exactly once, forming a closed loop

Q30ai) Hamilton Circuit :- $V_0 - V_1 - V_5 - V_4 - V_7 - V_6 - V_2 - V_3 - V_0$

Hamilton Path :- $V_0 - V_1 - V_5 - V_4 - V_7 - V_6 - V_2 - V_3$

have/contain

Note :- (As this graph is an Hamilton circuit, it is obvious it will have paths as well)

aii) Hamilton Circuit :- DOES NOT Exist (reason being that there is no closed loop formed that returns us back to the starting vertex, without repeating other vertices)

Hamilton Path :- b-a-e-h-g-f-c-d

iii) Hamilton Circuit:- $a-d-g-f-e-c-b-a$

Hamilton Path:- $a-d-g-f-e-c-b$

Note:- (if a graph is hamilton circuit than it will ~~have~~ contain hamilton path as well) (so we derive path by removing starting edge from end).

Q30bi) Hamilton circuit: $a-b-c-d-e-f-g-h-a$

Hamilton path:- $a-h-b-c-d-e-f-g$

bii) Hamilton Circuit: DOES NOT Exist (reason being ^{that} no the is no closed loop formed that return us back to the starting vertex, without repeating other vertices).

Hamilton Path:- $d-e-f-g-j-c-b-a-h$

biii) Hamilton Circuit:- $A-B-C-D-E-F-A$

Hamilton Path:- $A-B-C-D-E-F$

Note (if a graph is hamilton circuit, than it will contain/have path as well).

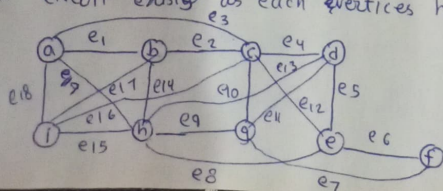
Q31ai) Euler circuit exists as each vertices have even degrees.

Route: $V_1 \xrightarrow{e_1} V_2 \xrightarrow{e_2} V_5 \xrightarrow{e_7} V_4 \xrightarrow{e_6} V_5 \xrightarrow{e_3} V_2 \xrightarrow{e_4} V_3 \xrightarrow{e_5} V_4 \xrightarrow{e_8} V_1$

all edges covered exactly once.

aii) Euler circuit does NOT exist as V_1, V_5, V_7, V_8 and V_4 have odd degrees. While the requirement for euler circuit was that all vertices should have even degrees.

aiii) Euler circuit exists as each vertices have even degrees



Total degree =
edges = $\frac{38}{2}$
= 19 edges

Route:- $a \xrightarrow{e_3} c \xrightarrow{e_4} d \xrightarrow{e_5} e \xrightarrow{e_6} f \xrightarrow{e_7} g \xrightarrow{e_{11}} d \xrightarrow{e_{13}} h \xrightarrow{e_9} g \xrightarrow{e_{10}} c \xrightarrow{e_{12}} e \xrightarrow{e_8} h$
 $\hookrightarrow a \xrightarrow{e_{18}} i \xrightarrow{e_{15}} h \xrightarrow{e_{14}} b \xrightarrow{e_{17}} i \xrightarrow{e_{16}} c \xrightarrow{e_2} b \xrightarrow{e_1} a$

as we can see all edges have been covered only once and we successfully reached our starting point.

Q31bi) Euler path does not exist as more than two vertices have odd degree.

bii) Euler path exists

Route: $U \xrightarrow{} V_1 \xrightarrow{} V_0 \xrightarrow{} V_7 \xrightarrow{} U \xrightarrow{} V_2 \xrightarrow{} V_4 \xrightarrow{} V_3 \xrightarrow{} V_2$
 $\hookrightarrow V_6 \xrightarrow{} V_4 \xrightarrow{} W \xrightarrow{} V_6 \xrightarrow{} V_5 \xrightarrow{} W$

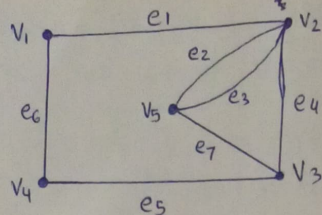
Q32ai) Incidence Matrix:-

	e_1	e_2	e_3	e_4	e_5	e_6	e_7
V_1	1	1	1	0	0	0	0
V_2	0	0	0	0	1	1	1
V_3	0	1	1	1	0	0	0
V_4	0	0	0	1	1	0	0
V_5	0	0	0	0	0	1	0
V_6	1	0	0	0	0	0	1

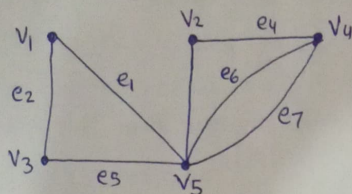
aii) Incidence Matrix:-

	e_1	e_2	e_3	e_4	e_5	e_6	e_7	e_8
V_1	1	1	1	0	0	0	0	0
V_2	0	1	1	1	0	1	1	0
V_3	0	0	0	1	1	0	0	0
V_4	0	0	0	0	0	0	1	1
V_5	0	0	0	0	1	1	0	0

Q32bi)



bii)



Q33ai) Adjacency List :

Initial Vertex	Terminal Vertices
a	a, b, c, d
b	d
c	a, b
d	b, c, d

Adjacency Matrix

a	1	1	1	1
b	0	0	0	1
c	1	1	0	0
d	0	1	1	1

aii) Adjacency List :

Initial Vertex	Terminal Vertices
a	b, d
b	a, c, d, e
c	b, c
d	a, e
e	c, e

Adjacency Matrix :

	a	b	c	d	e
a	0	1	0	1	0
b	1	0	1	1	1
c	0	1	1	0	0
d	1	0	0	0	1
e	0	0	1	0	1

aiii) Adjacency List :

Initial Vertex	Terminal Vertices
a	b, d
b	a, c
c	b, d
d	a, c

Adjacency Matrix

	a	b	c	d
a	0	3	0	1
b	3	0	1	0
c	0	1	0	3
d	1	0	3	0

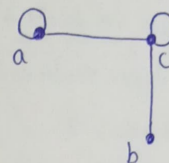
Q33aii) Adjacency List

Initial Vertex	Terminal Vertex
a	a, c, d
b	b, c, d
c	a, b, c
d	a, c, d

Adjacency Matrix

1	0	2	1
0	1	1	2
2	1	1	0
1	2	0	1

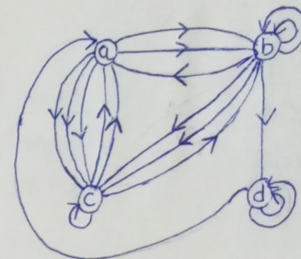
Q33bi)



for $\begin{bmatrix} a & b & c \\ 1 & 0 & 1 \\ 0 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

bii)

0	2	3	0
1	2	2	1
2	1	1	0
1	0	0	2



Q34ai) The graph is planar: Redrawn $\rightarrow$

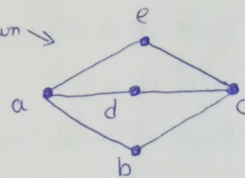
Vertices = 5
edge = 6

Euler formula: $V \geq 3$ so $e \leq 3V - 6$

$e \leq 3(5) - 6$

$e \leq 9$

True $\therefore$ planar



aii) The graph is not planar. (It's a different style of $K_{3,3}$)

vertices = 6

edge = 9

Euler formula: $V \geq 3$ so $e \leq 2V - 4$ (collary 3)

$e \leq 2(6) - 4$ $e \leq 8$

$e \leq 12 - 4$ $e \leq 8$

and as our edge is 9

$\therefore$ it is not planar. Also this graph is of $K_{3,3}$ hence it's bipartite graph and not planar

Q34b) For Graph G

Vertices = 4

Edges = 6

Region = $3 + 1 = 4$

For graph H

vertices = 6

edges = 9

Region = $4 + 1 = 5$

For graph J

Vertices = 5

edges = 10

region = $6 + 1 = 7$

Euler Formula verification:

Region + Edges + Vertices = 2

$$4 + 6 + 4 = 2$$

$$2 = 2 \text{ proof}$$

Euler Formula verification:

Region - Edges + Vertices = 2

$$5 - 9 + 6 = 2$$

$$2 = 2 \text{ proof}$$

Euler Formula verification:

Region - Edges + Vertices = 2

$$7 - 10 + 5 = 2$$

$$2 = 2 \text{ proof}$$

Q35a) Level of $n = 3$

b) Level of $a = 0$

c) Height of rooted tree = 5

d) Children of $n = u, v$

e) Parent of $g = d$

f) Siblings of $j = k$ and l

g) Descendants of $f = m, s, t, x, y$

h) Internal nodes = $a, b, c, d, e, f, h, k, m, n, o, t, v$

i) Ancestors of $z = v, n, h, d, a$

j) Leaves = $j, l, q, r, s, x, y, g, u, z, w, p$

Q36ai)

$$\{v_0, v_5\} = 4$$

$$\{v_5, v_6\} = 8$$

$$\{v_6, v_7\} = 13$$

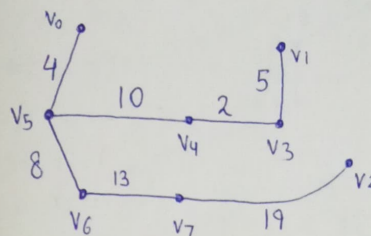
$$\{v_7, v_2\} = 19$$

$$\{v_5, v_4\} = 10$$

$$\{v_4, v_3\} = 2$$

$$\{v_3, v_1\} = 5$$

Total MST cost = 61



aii)

$$\{v_0, v_1\} = 4$$

~~$$\{v_1, v_2\} = 8$$~~

$$\{v_3, v_2\} = 2$$

$$\{v_3, v_4\} = 7$$

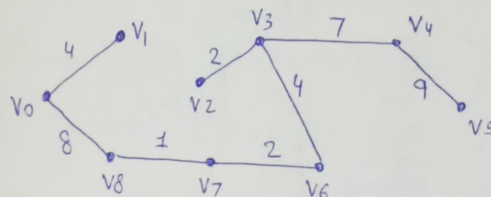
$$\{v_0, v_8\} = 8$$

$$\{v_8, v_7\} = 1$$

$$\{v_7, v_6\} = 2$$

$$\{v_4, v_5\} = 9$$

Total MST = 37



Q36bi)

$$\{d, f\} = 1$$

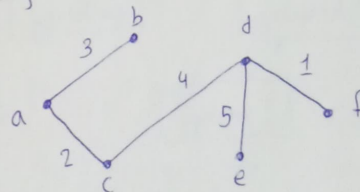
$$\{c, d\} = 4$$

$$\{a, c\} = 2$$

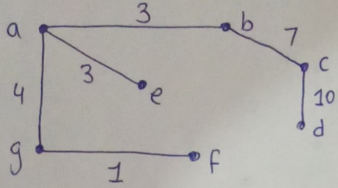
$$\{d, e\} = 5$$

$$\{a, b\} = 3$$

Minimum spanning = 15

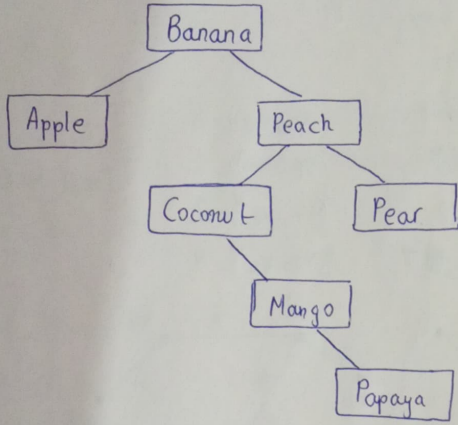


bii) $\{a, b\} = 3$ $\{g, f\} = 1$
 $\{a, e\} = 3$ $\{b, c\} = 7$
 $\{a, g\} = 4$ $\{c, d\} = 10$

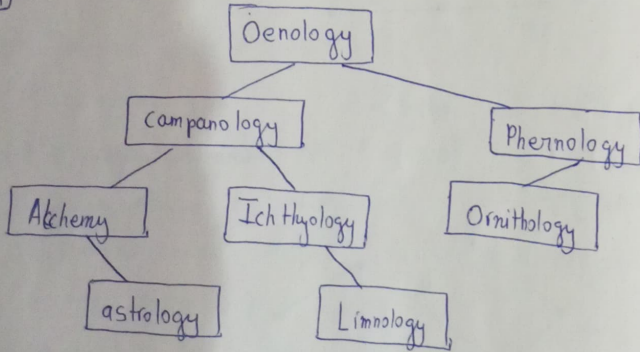


Minimum spanning = 28

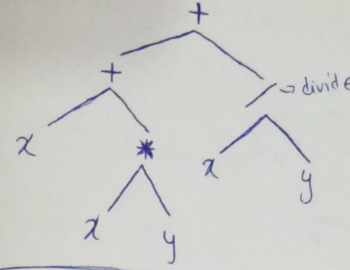
Q37ai)



ii)

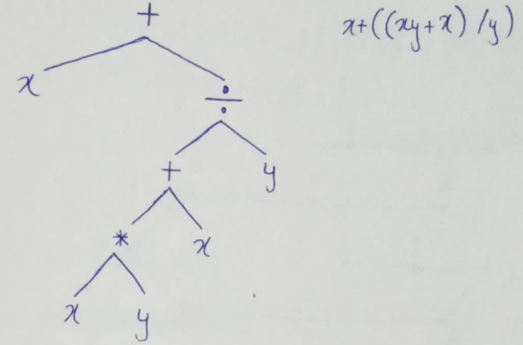


Q37bi)



$(x + xy) + (x/y)$

ii)



Q38ai) Pre-order = abeklmfgrscdhoijpq

Post-order = klme frsngbc oh ipqjda

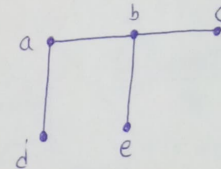
In-order = kelmbfrnsgacohdipjq

aii) Pre-order = abeflmncghopdijqsturk

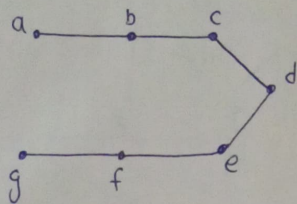
Post-order = elmnfbgophc isutq rjkda

In-order = eblfmnagcohpid squtjrle

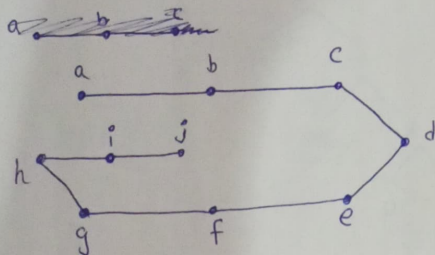
bi) Removing $\{b, d\}$, $\{e, c\}$



38bii) Removing: $\{a,d\}, \{b,g\}, \{b,e\}, \{g,e\}, \{g,d\}, \{a,d\}, \{b,f\}, \{c,e\}$



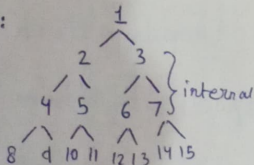
biii) Removing: $\{a,i\}, \{b,i\}, \{b,j\}, \{c,j\}, \{d,j\}, \{e,j\}, \{f,j\}, \{f,i\}, \{g,i\}$



Q39a) Edge = Vertex - 1 so 999 edges for 1000 vertices in tree as $n-1$ edge produces n -vertices.

b) full binary tree with 1000 internal vertices have 2000 edges. As each internal vertex have 2 edge, ~~and including root node edge + 1~~

eg:



we have 6 internal vertex who have 12 ~~nodes~~ edges

Applying formula $n = mi + 1$

$$n = 5(100) + 1$$

$$n = 501 \text{ vertices}$$

$$\text{Leaves} = \text{Vertex} - \frac{(\text{Vertex} - 1)}{m}$$

$$\text{Leaves} = 100 - \frac{(100 - 1)}{3} = 67 \text{ leaves}$$

$m = \text{no. of children each node has}$

$i = \text{no. of internal vertices}$

where m is tree type

Total Vertice = $L + i$

$$= (m-1)i + 1 + i$$

$$= (5-1)(100) + 1 + 100$$

$$= 501 \text{ vertices}$$

where Leaves $(L) = \frac{(m-1)i + 1}{1}$
 $= (m-1)i + 1$

39d) Leaves = $\frac{(m-1)i + 1}{1}$ for i $V = (m-1)i + 1 + i$

$$\text{Leaves} = (m-1)\left(\frac{V-1}{m}\right) + 1$$

$$\text{Leaves} = (3-1)\left(\frac{100-1}{3}\right) + 1$$

$$\text{Leaves} = 2(33) + 1$$

$$\text{Leaves} = 67 \text{ Leaves Answer}$$

Q40a) Prefix = $++x * xy / xy$

Postfix = ~~$xxy * xxy / xxy * xxy /$~~ $xxy * + xy / +$

ii) Prefix = $+x / + * xy xy$

Postfix = $xxy * x + y / +$

bi) Prefix = $+ \uparrow 32 \uparrow 23 / 6 - 42$

sol

$$= (3^2, -2^3) + (6 / (4-2))$$

$$= 9 - 8 + (6/2)$$

$$= 9 + 3$$

$$= 12$$

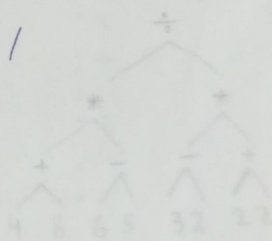
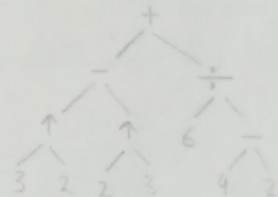
ii) Postfix = $48 + 65 - * 32 - 22 + * /$

sol: $((4+8) * (6-5)) / ((3-2) * (2+2))$

$$= (12 * (1)) / (1 * (4))$$

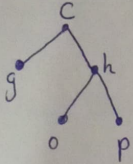
$$= 12/4$$

$$= 3 \text{ sol}$$

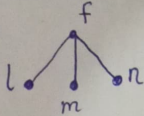


- Q4a) The rooted tree is not a full m-ary tree because some of the internal vertices have 2 children while others have 3 children.
- b) It is not a balanced tree because it has leaves at levels 2, 3, 4 and 5.

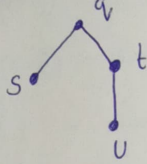
c) Root c



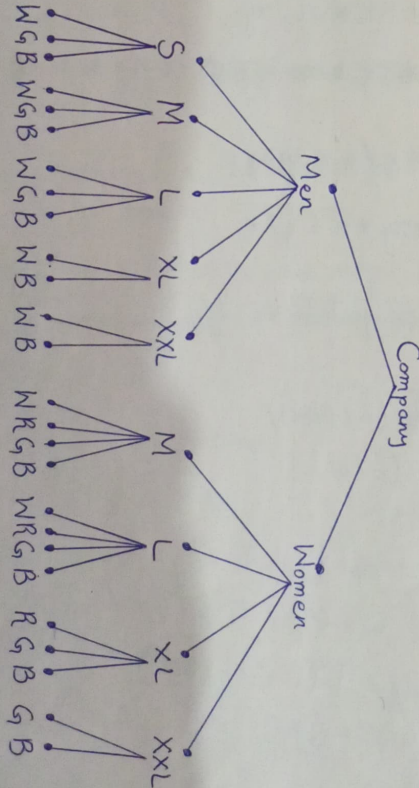
ii) Root f



iii) Root q



Q4b)



W - white
R - red
G - Green
B - Black

Minimum number of T-shirts = 26

working

$$\text{women} = 4 + 4 + 3 + 2 = 13 - \text{tshirt}$$

$$\text{men} = 3 + 3 + 3 + 2 + 2 = 13 - \text{tshirt}$$

$$\text{total} = 13 + 13 = 26 \text{ tshirt}$$

b) Internal Vertices (nodes that have children) = ?

working:

Level 0: root (1 node)

Level 1: women, men (2 nodes)

Level 2: M, L, XL, XXL (women) + S, M, L, XL, XXL (men) = 9 nodes

$$\text{Total Internal Vertices} = 1 + 2 + 9 = 12$$

Height of tree = 3

Leaves = 26

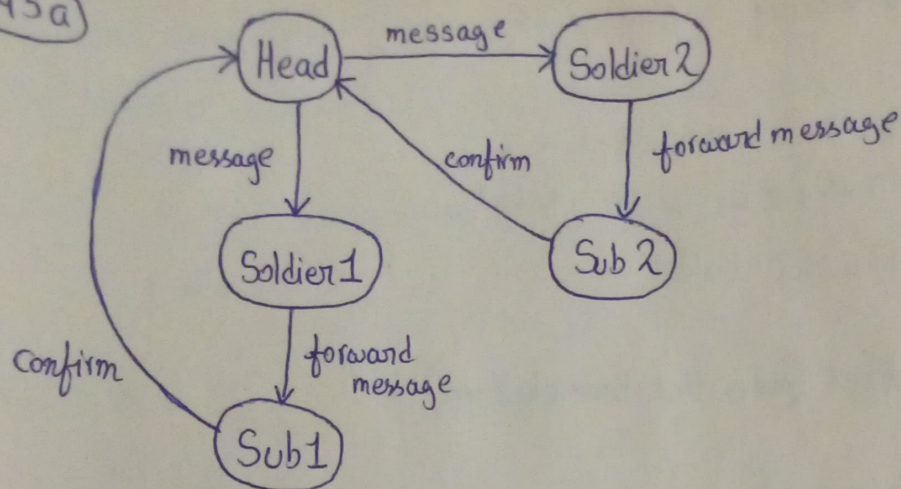
i) No it's not a full m-ary tree as some nodes have 2, 4, 5, 3 children and so on, (not equal children)

ii) Yes it is a balanced tree for all leaves are at the same level that is Level 3. (all paths from root to leaves have the same length that is 3)

i) Pre order :- Men, S, W, G, B, M, W, G, B, L, W, G, B, XL, W, B, XL, W, B, Women, M, W, R, G, B, L, W, R, G, B, XL, R, G, B, XXL, G, B

Post order :- W, G, B, S, W, G, B, M, W, G, B, L, W, B, XL, W, B, XXL, Men, W, R, G, B, M, W, R, G, B, L, R, G, B, XL, G, B, XXL, Women, Company

Q43a)



Vertex	In-degree	Out-degree
Head	2	2
Soldier 1	1	1
Sub 1	1	1
Soldier 2	1	1
Sub 2	1	1

Ex

Q44a)

N	d(a)	d(b)	d(c)	d(d)	d(e)	d(f)	d(S)
H	3, H	2, H	5, H	∞	∞	∞	∞
Hb	3, H		5, H	3, b	8, b	∞	∞
Hbd	3, H		5, H		8, b	7, d	∞
Hbda			5, H		8, b	7, d	∞
Hbdac					7, c	7, d	∞
Hbdace						7, d	11, e
Hbdacef							9, f
HbdacefS							

H = home
S = school

shortest path = ~~H~~ H b d f S (9)

44b) $\deg(H) = 3$ $\deg(d) = 3$
 $\deg(a) = 2$ $\deg(e) = 4$
 $\deg(b) = 3$ $\deg(f) = 3$
 $\deg(c) = 2$ $\deg(S) = 2$

We can not have Euler Circuit as we have vertices with odd degrees, and the requirement for the circuit is vertices to be of even degree.

We can't have Euler Path as well because for a path we must have only 2 vertices with odd degree but we have 4 vertices of odd-degree (including home) hence can't make euler path

44c) It does not have a hamilton circuit. It failed both Dirac's theorem (as per him not all of our vertices have degree ~~greater~~ greater than equal to 4)

working Dirac's theorem

vertex $\hookleftarrow n \geq 3$ then degree of every vertex $\geq n/2$ $8/2 = 4$
 $8 \geq 3$ condition failed

vertex H, a, b, c, d, f, S are less than 4

It also doesn't satisfy Ore's theorem.

If we try to brute force a path from House to school, there are vertices that remain unvisited or are traversed multiple times.

The graph doesn't have a Hamilton path if we start from House to School. As we have vertex C left unvisited

$H \rightarrow A \rightarrow D \rightarrow B \rightarrow E \rightarrow F \rightarrow \text{School}$

However if we start from A then we have a path

$A \rightarrow H \rightarrow C \rightarrow E \rightarrow B \rightarrow D \rightarrow F \rightarrow S$
 home school

Q45a) Yes it is isomorphic

Proof

Vertices (Figure 1) = 8

edge = 12

degree sequence = 3, 3, 3, 3, 3, 3, 3, 3

A B C D W X Y Z

vertices (Figure 2) = 8

edge = 12

degree sequence:

$\Rightarrow 3, 3, 3, 3, 3, 3, 3, 3$

1 2 3 4 5 6 7 8

As qty of vertex, edge, degree matches

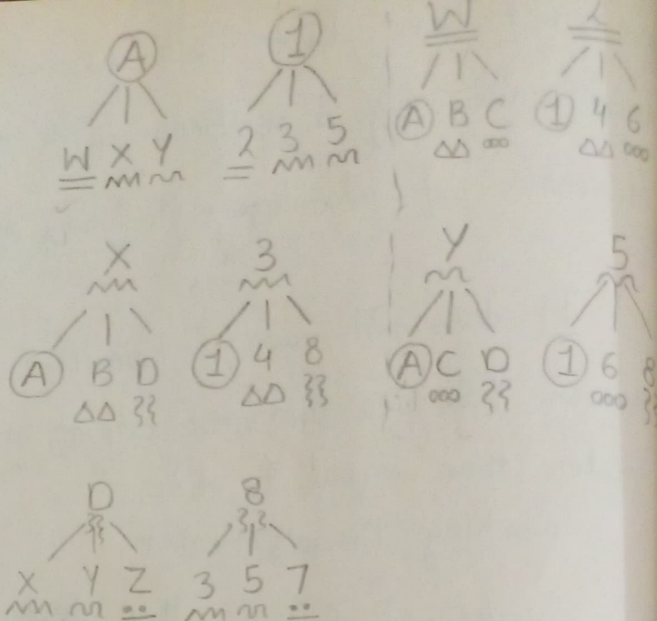
Mapping : $V(G) \rightarrow V(H)$

$$A = 1 \quad W = 2$$

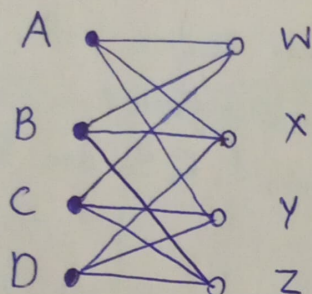
$$B = 4 \quad X = 3$$

$$C = 6 \quad Y = 5$$

$$D = 8 \quad Z = 7$$



bi)



$$\text{set-left} = \{A, B, C, D\}$$

$$\text{set-right} = \{W, X, Y, Z\}$$

Yes graph G is a bipartite, it passes color theory test as the adjacent nodes connected by edges are of opposite color (blue, white). Moreover the set from left side is connected to the set of right with no edges joining the vertices belonging from the same set.

ii) Euler circuit doesn't exist as each vertex have odd degrees.
Hamilton circuit exists

Route :- $1 \rightarrow 2 \rightarrow 4 \rightarrow 7 \rightarrow 6 \rightarrow 5 \rightarrow 8 \rightarrow 3 \rightarrow 1$

Q46ai) $f(a)=b$, $f(b)=a$, $f(c)=c$, $f(d)=d$

Domain = $\{a, b, c, d\}$

Co-domain = $\{a, b, c, d\}$

Range = $\{a, b, c, d\}$

aii) Injective ? Yes (all image are distinct)

Surjective ? Yes (range = codomain)

Therefore bijective (as both injective and surjective)

iii) Inverse exists (since bijection)

$$f^{-1}(b) = a, f^{-1}(a) = b, f^{-1}(c) = c, f^{-1}(d) = d$$

bi) $f(a) = b, f(b) = b, f(c) = d, f(d) = c$

$$\text{Domain} = \{a, b, c, d\}$$

$$\text{Co-domain} = \{a, b, c, d\}$$

$$\text{Range} = \{b, c, d\}$$

ii) Injective ? No (both a and b map to b)

Surjective ? No (element a $\in B$ is not hit)

Therefore not bijective

iii) Inverse does not exist (as it's not one to one or onto)

ci) $f(a) = d, f(b) = b, f(c) = c, f(d) = d$

$$\text{Domain} = \{a, b, c, d\}$$

$$\text{co-domain} = \{a, b, c, d\}$$

$$\text{range} = \{b, c, d\}$$

ii) Injective ? No (both a and d map to d)

Surjective ? No (element a $\in B$ is not hit)

Therefore not bijective

iii) Inverse does not exist (as it's not one to one or onto)

di) $f(a) = c, f(b) = a, f(c) = b, f(d) = d$

$$\text{Domain} = \{a, b, c, d\}$$

$$\text{co-domain} = \{a, b, c, d\}$$

$$\text{Range} = \{a, b, c, d\}$$

ii) Injective ? Yes (images are all distinct)

Surjective ? Yes (range = ~~set~~ co-domain)

Therefore bijective (as both injective and surjective)

iii) Inverse exists (since bijective)

$$f^{-1}(c)=a, f^{-1}(a)=b, f^{-1}(b)=c, f^{-1}(d)=d$$

47a
Q 28a) $f(x) = \left\lfloor \frac{x^2}{3} \right\rfloor$ a floor function

i) $S = \{-2, -1, 0, 1, 2, 3\}$

$$f(-2) = \left\lfloor \frac{4}{3} \right\rfloor = 1 \quad \left| \quad f(-1) = \left\lfloor \frac{1}{3} \right\rfloor = 0 \quad \left| \quad f(0) = \left\lfloor \frac{0}{3} \right\rfloor = 0$$

$$f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0 \quad \left| \quad f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1 \quad \left| \quad f(3) = \left\lfloor \frac{9}{3} \right\rfloor = 3$$

$$\therefore f(S) = \{1, 0, 0, 0, 1, 3\} \quad \text{Answer}$$

aii) $S = \{0, 1, 2, 3, 4, 5\}$

$$f(0) = \left\lfloor \frac{0}{3} \right\rfloor = 0 \quad \left| \quad f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0 \quad \left| \quad f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1$$

$$f(3) = \left\lfloor \frac{9}{3} \right\rfloor = 3 \quad \left| \quad f(4) = \left\lfloor \frac{16}{3} \right\rfloor = 5 \quad \left| \quad f(5) = \left\lfloor \frac{25}{3} \right\rfloor = 8$$

$$\therefore f(S) = \{0, 0, 1, 3, 5, 8\} \quad \text{Answer}$$

aiii) $S = \{1, 5, 7, 11\}$

$$f(1) = \left\lfloor \frac{1}{3} \right\rfloor = 0 \quad \left| \quad f(5) = \left\lfloor \frac{25}{3} \right\rfloor = 8 \quad \left| \quad f(7) = \left\lfloor \frac{49}{3} \right\rfloor = 16 \quad \left| \quad f(11) = \left\lfloor \frac{121}{3} \right\rfloor = 40$$

$$\therefore f(S) = \{0, 8, 16, 40\} \quad \text{Answer}$$

aiiv) $S = \{2, 6, 10, 14\}$

$$f(2) = \left\lfloor \frac{4}{3} \right\rfloor = 1 \quad \left| \quad f(6) = \left\lfloor \frac{36}{3} \right\rfloor = 12 \quad \left| \quad f(10) = \left\lfloor \frac{100}{3} \right\rfloor = 33 \quad \left| \quad f(14) = \left\lfloor \frac{196}{3} \right\rfloor = 65$$

$$\therefore f(S) = \{1, 12, 33, 65\} \quad \text{Answer}$$

47bi) $\lceil \frac{3}{4} \rceil = 1$ ceiling function

bii) $\lfloor \frac{7}{8} \rfloor = 0$ floor function

biii) $\lceil -\frac{3}{4} \rceil = 0$ ceiling function

biv) $\lfloor -\frac{7}{8} \rfloor = -1$ floor function

bv) $\lceil 3 \rceil = 3$ ceiling function

bvi) $\lfloor -1 \rfloor = -1$ floor function

bvii) $\lfloor \frac{1}{2} + \lceil \frac{3}{2} \rceil \rfloor = \lfloor \frac{1}{2} + 2 \rfloor = 2$ floor (ceiling) function

bviii) $\lfloor \frac{1}{2} \lfloor \frac{5}{2} \rfloor \rfloor = \lfloor \frac{1}{2} (2) \rfloor = 1$ floor (floor) function

Q47c) sol

let $x = \frac{3}{4}$

~~$\frac{3}{4}$~~ $\lfloor -x \rfloor = -\lceil x \rceil$

$\lfloor -\frac{3}{4} \rfloor = -\lceil \frac{3}{4} \rceil$

$-1 = -1$

let $x = \frac{3}{4}$

$\lceil -x \rceil = -\lfloor x \rfloor$

$\lceil -\frac{3}{4} \rceil = -\lfloor \frac{3}{4} \rfloor$

$0 = 0$

LHS = RHS
proof

LHS = RHS proof

As both the statement result match of LHS and RHS.

Therefore it's a proof

Q48a)

~~Q48a)~~ $f(a) = 2a + 3$

$g(a) = 3a + 2$

Sol

composition of f and g

$(fg)(a) = f(g(a)) = 2(3a + 2) + 3$

$= 6a + 4 + 3$

$= 6a + 7$ Ans

composition of g and f

$$\begin{aligned}(gf)(a) &= g(f(a)) = 3(2a+3)+2 \\ &= 6a+9+2 \\ &= 6a+11 \quad \text{Answer}\end{aligned}$$

~~248~~ 48b) Both f and g are linear function of the ~~map~~ form $ma+c$. f, g are injective but not surjective

48c) f and g are not invertible as they are not bijective cause they weren't surjective

Q49a) Functions

Application 1: Machine Learning (Predictions)

A trained ML model is essentially a function mapping input data to outputs. They help predict results from data, measure how wrong or right the predictions are (cost and loss functions), and train models to improve accuracy.

Application 2: Algorithm design

Functions describe how an algorithm transforms input into output. eg sorting (a function takes an unsorted list and outputs a sorted one), searching (a function maps a key to its position in data).

b) Relations

Application 1: Social Networks

In platforms like Facebook or Instagram, the friendship or follow connection between two users is a real-world example of a relation, often represented as a set of ordered pairs (User A, User B).

Application 2: Databases

Relation databases (like SQL), where tables represent relationship between entities, for example linking "Student_ID" in a students table with the same "Student_ID" in a grades table

c) Sequence

Application 1: Data Structures (Arrays and Lists)

These data structures store elements in sequential order, which is crucial for algorithm design, sorting and searching operations.

Application 2: Algorithmic Analysis

It helps express time complexity and performance patterns. For eg recurrence relations like $T(n) = 2T(\frac{n}{2}) + n$ describes the sequence of steps in recursive algorithm such as merge sort.

d) Series:

Application 1: Algorithm efficiency and computation

Series are used to evaluate the total number of operations an algorithm performs, especially when analyzing loops or recursive calls.

Application 2: Computer Graphics and Signal Processing

Fourier series are used to represent images, sounds, and signals digitally allowing compression and reconstruction in multimedia applications.

e) Graph Theory

Application 1: Social Networks and Web Structure

Webpages and social networks can be modeled as graphs, where nodes represent user or pages, and edges presents links or friendship.

Application 2: Networking and Routing

Graph algorithms (like Dijkstra's) are used in network routing to find the shortest or most efficient path between devices in a computer network.

f) Trees

Application 1: Data Representation and Storage

Trees form the backbone data structures like Binary Search Tree, Heaps, and tries, which are used in compilers, search engines and

databases for efficient data lookup and sorting.

Application 2 :- Decision Making and AI

Decision trees and hierarchical clustering use tree structures to classify data, make predictions, and assist in machine learning models.